Social Media Analytics

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1. Introduction

We take a look at data of 1.6 million twitter users and draw useful insights while exploring interesting patterns. The techniques used include Text mining, sentimental analysis, probability, building a time series data from the existing data set and Hierarchical clustering on text/words.

1.1 Data Description

We use two different files in our data sets:

1. The *tweets.csv* data set contains 1.6 million tweets with 6 fields as follows:

- target: the polarity of the tweet (0 = negative, 2 = neutral, 4 = positive)
- ids: The id of the tweet (2087)
- date: the date of the tweet (Sat May 16 23:58:44 UTC 2009)
- flag: The query (lyx). If there is no query, then this value is NO_QUERY.
- user: the user that tweeted (robotickilldozr)
- text: the text of the tweet (Lyx is cool)

2. The *daily-website-visitors.csv* contains 5 years of daily time series data for several measures of traffic with 2167 records and 8 Columns:

- Row: Unique row number for each record
- Day: Day of week in text fomr (Sunday, Monday, etc)
- Day.Of.Week: (Day of week in numeric form (1-7))
- Date: Date in mm/dd/yyyy format
- Page.Loads: Daily number of pages loaded
- Unique.Visits: Daily number of visitors from whose IP addresses there haven't been hits on any page in over 6 hours
- First.Time.Visits: Number of unique visitors who do not have a cookie identifying them as a previous customer
- Returning.Visits: Number of unique visitors minus first time visitors

1.2 Data Acquisition

We acquire bot the data sets from Kaggle:

1. https://www.kaggle.com/kazanova/sentiment140

2. https://www.kaggle.com/bobnau/daily-website-visitors

Table 1: Previewing few columns of Twitter user data set

date	text
Mon Apr 06 22:19:45 PDT 2009	@switchfoot http://twitpic.com/2y1zl - Awww, that's a bummer. You shoulda got David Carr of Third Day to do it. ;D
Mon Apr 06 22:19:49 PDT 2009	is upset that he can't update his Facebook by texting it... and might cry as a result School today also. Blah!
Mon Apr 06 22:19:53 PDT 2009	@Kenichan I dived many times for the ball. Managed to save 50% The rest go out of bounds
Mon Apr 06 22:19:57 PDT 2009	my whole body feels itchy and like its on fire
Mon Apr 06 22:19:57 PDT 2009	@nationwideclass no, it's not behaving at all. i'm mad. why am i here? because I can't see you all over there.

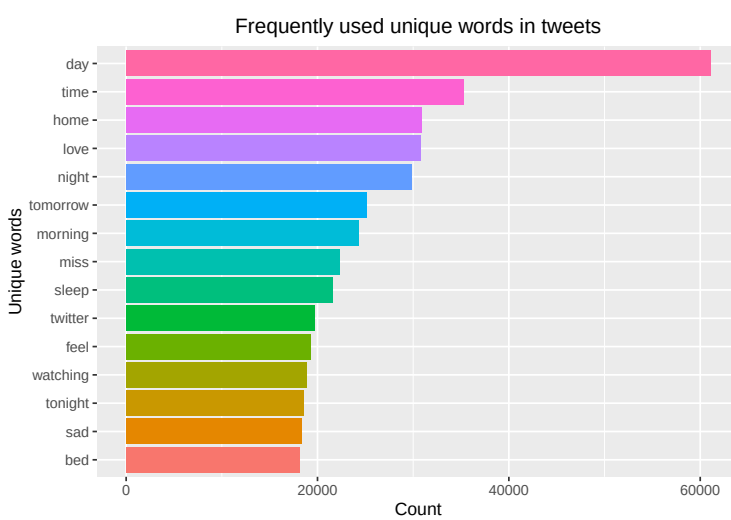
Table 2: Previewing few columns of Daily time series data set.

Row	Day	Date	Page.Loads	Unique.Visits
1	Sunday	9/14/2014	2,146	1,582
2	Monday	9/15/2014	3,621	2,528
3	Tuesday	9/16/2014	3,698	2,630
4	Wednesday	9/17/2014	3,667	2,614
5	Thursday	9/18/2014	3,316	2,366

2.Analytical Questions

2.1 Text Mining

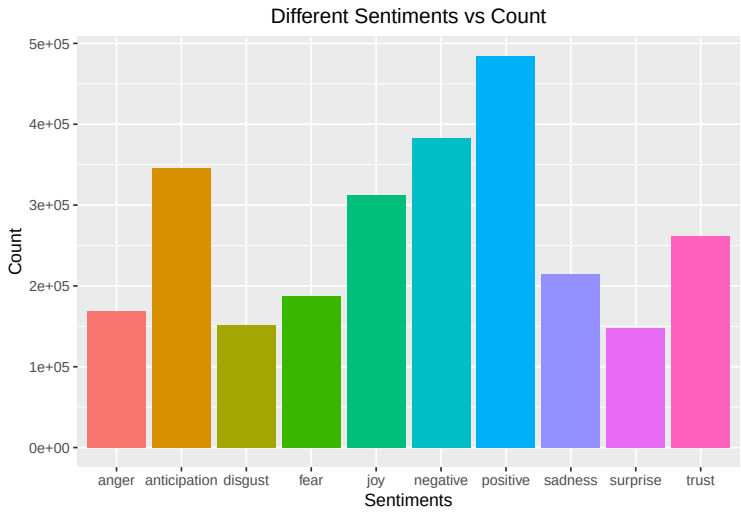
2.1.1 Finding the frequently used unique words



For this insight, we consider only the *original* thought of the user/author. We Remove stop words, username mentions, replies, and Re-tweets so that we only have the "original" tweets and visualize our findings.

Observation: *Day* is the most frequently used word which has been used around 63,000 times out of the total of 1.6 million tweets. Following that, the words *Time*, *Home*, *love* and *night* have been used around 30,000 times each.

2.1.2 Sentimental Trends of Tweets

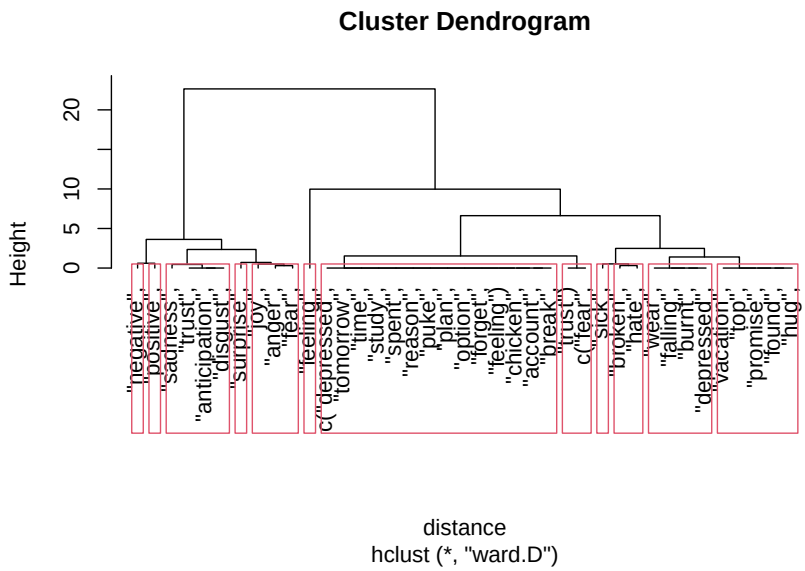


By utilizing the nrc library, we find different sentiments in each of the tweets and visualize their counts.

Observation: *Positive, negative, anticipation* are the top three most tweeted sentiments. Another trend is that there are equal number of *Anger, disgust and surprise* sentiment tweets. A lot of Users have also tweeted about issues that they *fear and trust.*

2.2 Clustering Analysis

Hierarchical clustering words by sentiments



Since our data set comprises of text data, we make a corpus and utilize Hierarchical clustering technique. This technique gives us a dendrogram of different words grouped together by sentiments. The number of clusters in hierarchical clustering is given as a range while trying to plot it. Using the suggested range, we can chose the number of clusters. We have chosen to go with 12 as the number of clusters.

Observation: The above Dendrogram clusters our sample space into 12 clusters grouped by sentiments. The height of the Dendrogram signifies the distance between the clusters.

2.3 Probability

2.3.1. Calculating the PMF and CDF

Table 3: First 5 records of PMF of the tweet frequency.

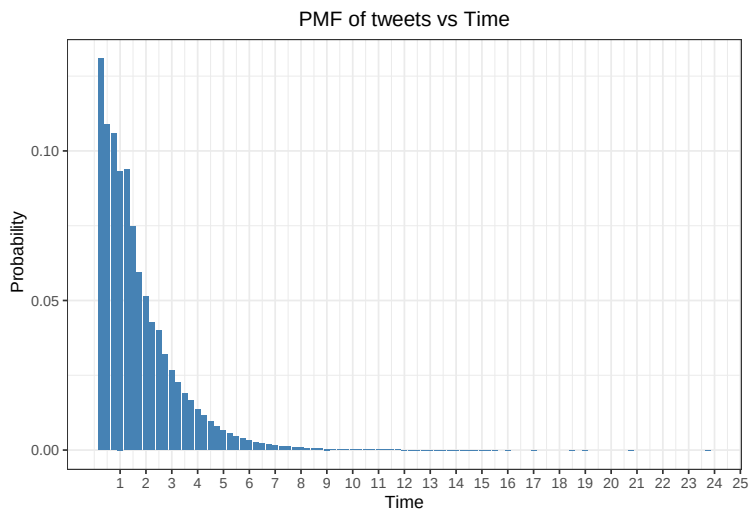
pickup_pmf
0.1307690
0.1087685
0.1058048
0.0932933
0.0937506

Table 4: First 5 records of CDF of the tweet frequency

pickup_cdf
0.1307690
0.2395376

pickup_cdf
0.3453424
0.4386357
0.5323863

2.3.2. Probability Mass Function over Time

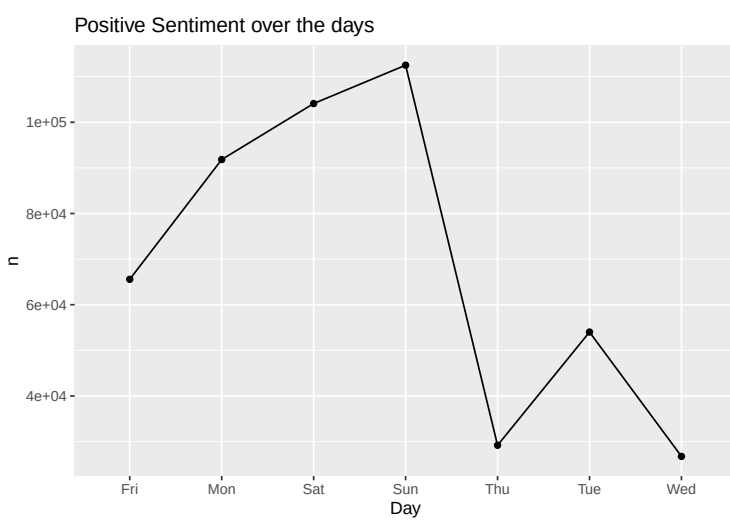


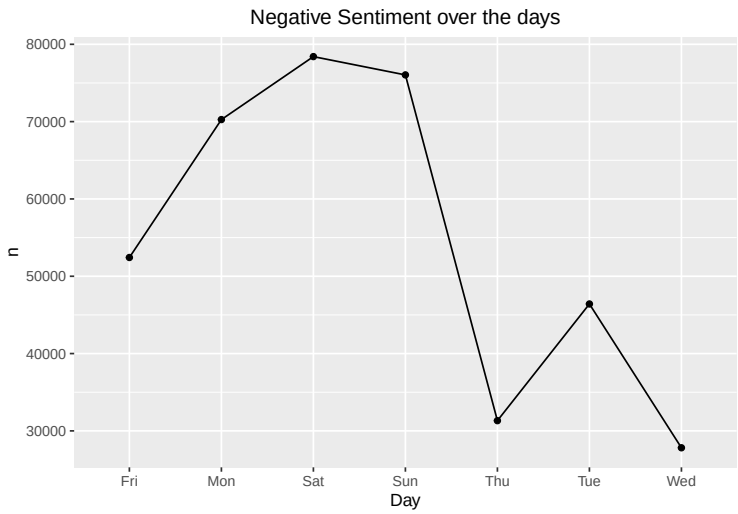
Observation: The probability of tweets is reducing over time in an exponential scale for a given period.The probability is highest in the start of the time chosen.

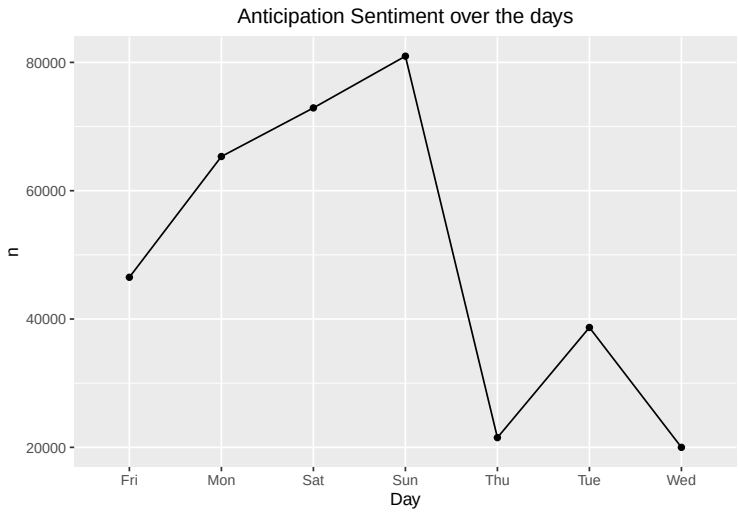
2.4 Time Series

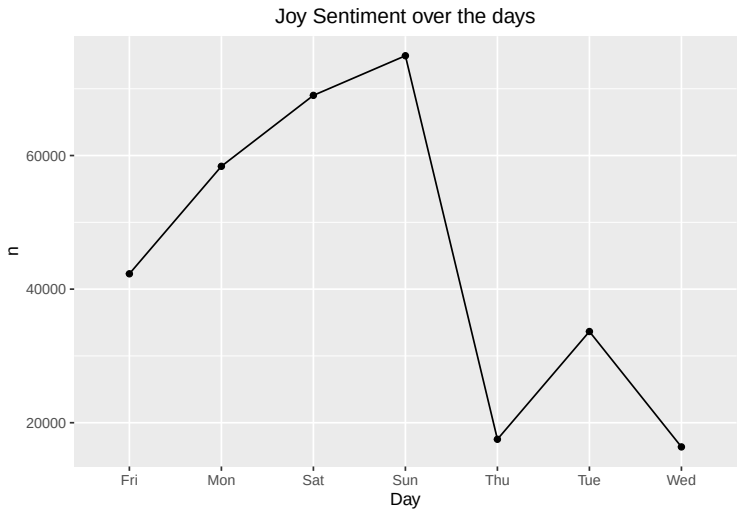
2.4.1. Trend analysis for different sentiments for each day of the week.

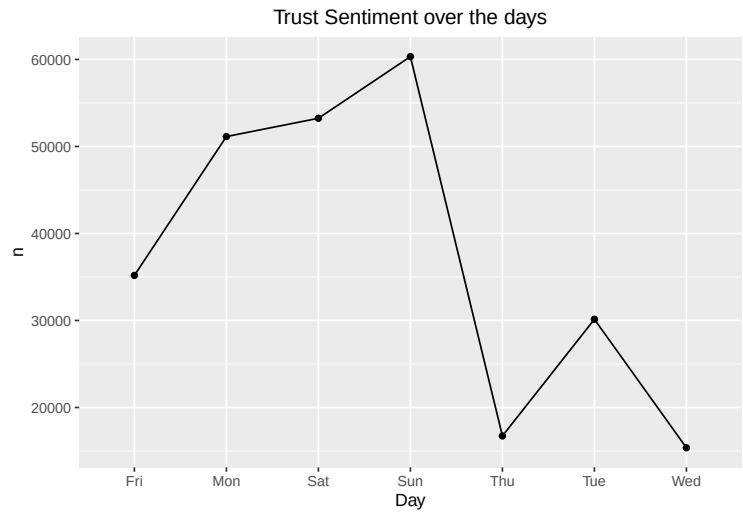
Extracting all the sentiments from the sentiments and date column to determine the sentiments related to each day. **All the counts related to the sentiments are mentioned in the original source code. We are reporting only the graphs for easier readability.**




Negative Sentiment over the days
80000
70000
60000
50000
40000
30000
n
Fri
Mon
Sat
Sun
Thu
Tue
Wed
Day

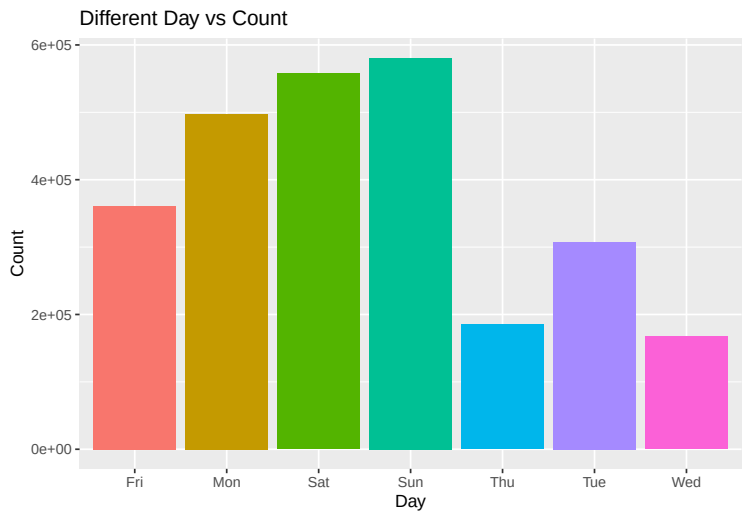

Anticipation Sentiment over the days
80000
60000
40000
20000
n
Fri
Mon
Sat
Sun
Thu
Tue
Wed
Day


Joy Sentiment over the days
60000
40000
20000
n
Fri
Mon
Sat
Sun
Thu
Tue
Wed
Day



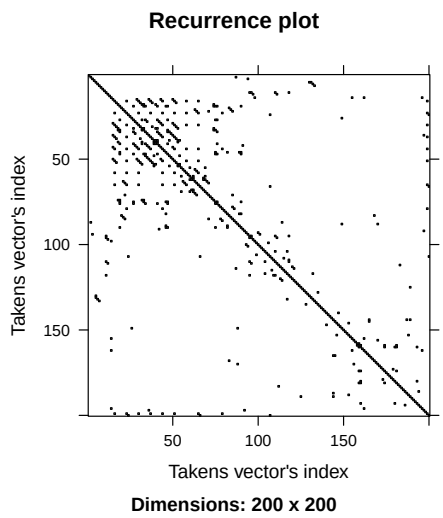
Observation: From all the above graphs, we observe that positive sentiments of tweets increases till sunday and then drastically decreases afterwards. The negative sentiments starts increasing till Saturday and then decreases.The other sentiments shown the graphs also follow a similar pattern to that of postive sentiment.

2.4.1 Trend analysis looking at number of tweets per day of the week



Observation: Top three days for tweeting are Saturday, Sunday and Monday which should be inline with the start of the weekend and the work week. Meanwhile Wednesday and Thursday have the lowest number of tweets as it's in the middle of the week.

RQA analysis



observation: From the RQA graph, it is observed that there are single isolated points. This can be interpreted as heavy fluctuation and the process may be an uncorrelated random or even anti-correlated process. Therefore, the number of page loads and the texts in the tweets are uncorrelated.

Degree of Permutation Entropy and Complexity

```
[16]
op_tday = ordinal_patterns(tday['count'],4,1)

print("Permutation Entropy tweet number =", p_entropy(op_tday))
print("Complexity tweet number=", complexity(op_tday))

Permutation Entropy tweet number = 0.9398649687446673
Complexity tweet number= 0.06763578543417548
```

Figure 1: The Permutation Entropy and Complexity of the tweet numbers

Observations: The permutation entropy is so high and the complexity near to zero. This means that there is no relationship between the dates and the day.

Number of Tweets per day over a period of 3 Months

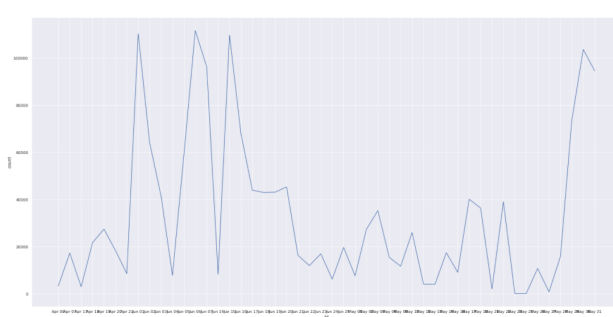


Figure 2: Number of Tweets over time

Observation: The trend of the number of tweets in three months. During may the trend changes dramatically and there are three highest number of tweets in may.

3. Summary

After careful analysis of 1.6 million worth of twitter data, We were able to decipher lot of emerging patterns and visualize them. We were able to gain valuable insights about our business questions through various plots,text analysis/mining, clustering, probability and time series data.

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